16-833: Robot Localization and Mapping, Spring 2026

Homework 2
SLAM using Extended Kalman Filter (EKF-SLAM)

Po-Lin Chen
polinc@andrew.cmu.edu

1 Theory

1.1 Next Pose Prediction

Given the current pose $p_t = [x_t, y_t, \theta_t]^\top$ and controls d_t, α_t, the next pose p_{t+1}:

$$p_{t+1} = g(p_t, u_t) = \begin{bmatrix} x_t + d_t \cos(\theta_t) \\ y_t + d_t \sin(\theta_t) \\ \theta_t + \alpha_t \end{bmatrix}$$

1.2 Predicted Uncertainty

Assuming zero-mean Gaussian errors e_x, e_y, e_α in the robot frame, let $R = \mathrm{diag}(\sigma_x^2, \sigma_y^2, \sigma_\alpha^2)$.

The predicted uncertainty Σ_{t+1} in the global frame:

$$\Sigma_{t+1} = G_t \Sigma_t G_t^\top + R_t$$

Where G_t is the Jacobian of g with respect to p_t:

$$G_t = \begin{bmatrix} 1 & 0 & -d_t \sin(\theta_t) \\ 0 & 1 & d_t \cos(\theta_t) \\ 0 & 0 & 1 \end{bmatrix}$$

And R_t is the process noise transformed to the global frame using H_t:

$$R_t = H_t R H_t^\top, \quad H_t = \begin{bmatrix} \cos(\theta_t) & -\sin(\theta_t) & 0 \\ \sin(\theta_t) & \cos(\theta_t) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.3 Estimated Landmark Position

Using range r and bearing β with respective noises n_r, n_β, the landmark position $l = (l_x, l_y)$ is estimated as:

$$\begin{bmatrix} l_x \\ l_y \end{bmatrix} = \begin{bmatrix} x_t + (r + n_r)\cos(\theta_t + \beta + n_\beta) \\ y_t + (r + n_r)\sin(\theta_t + \beta + n_\beta) \end{bmatrix}$$

1.4 Predicted Measurement

Given $l = (l_x, l_y)$ and pose p_t, define $\delta_x = (l_x - x_t)$ and $\delta_y = (l_y - y_t)$. The predicted measurements are:

$$\begin{bmatrix} \beta \\ r \end{bmatrix} = \begin{bmatrix} \text{wrap2pi}(\text{atan2}(\delta_y, \delta_x) - \theta_t) + n_\beta \\ \sqrt{\delta_x^2 + \delta_y^2} + n_r \end{bmatrix}$$

1.5 Measurement Jacobian H_p

Define $h = \begin{bmatrix} \beta \\ r \end{bmatrix}$, $q = \delta_x^2 + \delta_y^2$. The Jacobian with respect to the robot pose p_t is:

$$H_p = \frac{\partial h}{\partial p_t} = \frac{1}{q} \begin{bmatrix} \delta_y & -\delta_x & -q \\ -\sqrt{q}\delta_x & -\sqrt{q}\delta_y & 0 \end{bmatrix}$$

1.6 Measurement Jacobian H_l

The Jacobian with respect to the landmark position l is:

$$H_l = \frac{\partial h}{\partial l} = \frac{1}{q} \begin{bmatrix} -\delta_y & \delta_x \\ \sqrt{q}\delta_x & \sqrt{q}\delta_y \end{bmatrix}$$

We do not need to calculate Jacobians for other landmarks because of the **conditional independence assumption**: the measurement of a specific landmark depends only on the current robot pose and that landmark's position, independent of other landmarks' states.

2 Implementation

2.1 Total landmarks

6 landmarks are observed in the entire sequence.

2.2 Trajectory

The trajectory at the final step is visualized in the Figure 1.

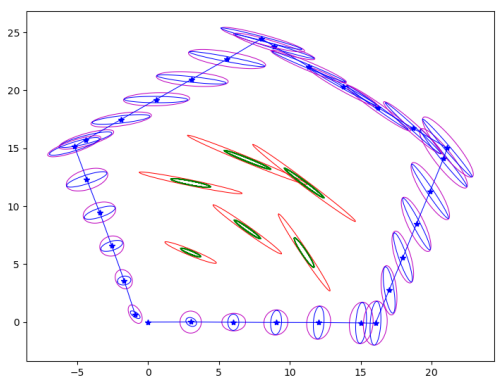


Figure 1: Trajectory

2.3 EKF-SLAM Analysis

The EKF-SLAM algorithm improves estimation by using the Kalman gain to weigh new measurements against the predicted state. While the robot's movement increases uncertainty (shown as larger magenta ellipses), observing landmarks provides new information that reduces the distribution size of both the robot (blue) and the landmarks (green).

2.4 Evaluation

All ground truth landmark positions fall within the smallest corresponding ellipses, indicating that the EKF-SLAM updates are correct and the estimated distributions are reliable. The small Euclidean and Mahalanobis distances confirm that the estimated positions are geometrically accurate and closely match the true probability distribution.

Table 1: Euclidean and Mahalanobis Distances for All Landmarks

Landmark #	**Euclidean**	**Mahalanobis**
1	0.003649	0.058384
2	0.004252	0.078725
3	0.004489	0.025582
4	0.004130	0.069339
5	0.002881	0.006024
6	0.003399	0.082529

3 Discussion

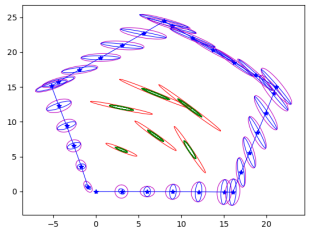
3.1 Non-Zeros in Covariance Matrix

Initially, the state covariance matrix is diagonal because landmarks are assumed to be independent of each other and the robot pose. During the update step, observing a landmark creates correlations between the robot and all landmarks through the Kalman gain. These statistical dependencies are captured as non-zero cross-covariance terms in the final matrix. The incorrect assumption made during initialization is that these state elements remain uncorrelated.

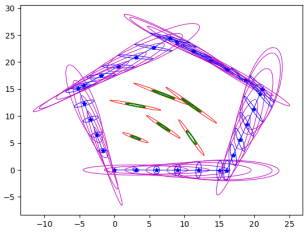
3.2 Uncertainty Parameters

Outputs with different uncertainty parameters are visualized in Figure 2.

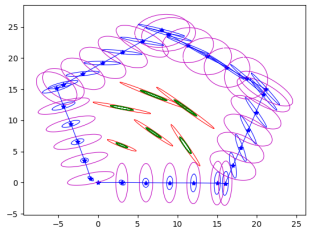
Figure 2. shows that **increasing σ_x by 10x (b)** leads to larger variance along x-axis, and similarly **increasing σ_y by 10x (c)** leads to larger variance along y-axis throughout the whole trajectory. As **increasing σ_α (d)**, only very subtle differences are observed, which suggests that this parameter may not be large to influence the performance. **Increasing σ_β (e)** amplifies landmark uncertainty perpendicularly to the robot's line of sight, and this angular ambiguity propagates through the state vector to elongate the predicted pose ellipses. **Increasing σ_r (f)** heavily affects the SLAM performance of the algorithm as it introduced extra landmark uncertainty and thus extra robot pose uncertainty.



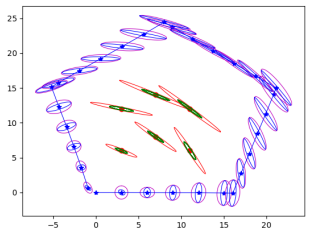
(a) Default Parameters



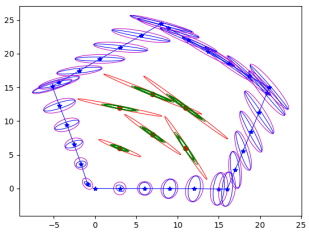
(b) Increased σ_x (10x)



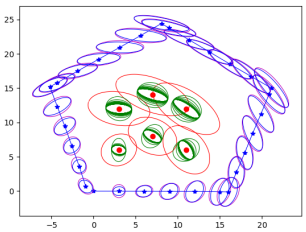
(c) Increased σ_y (10x)



(d) Increased σ_α (10x)



(e) Increased σ_β (10x)



(f) Increased σ_r (10x)

Figure 2: Sensitivity Analysis of EKF-SLAM: Comparison of robot trajectory and landmark estimation under different uncertainty parameters. Increased motion noise $(\sigma_x, \sigma_y, \sigma_\alpha)$ primarily affects the robot pose ellipses, while increased measurement noise (σ_β, σ_r) expands the landmark and pose uncertainty in specific directions.

3.3 Computation Optimizations

- **Submapping**: Divide the environment into local submaps that are solved independently and stitched together offline, ensuring a fixed number of variables n are updated per step.
- **Sparse Matrix**: Utilize sparse matrix representations and optimal factorization to reduce complexity of matrix inversion during the update step.
- **Selective State Updates**: Update only the state variables involved in current measurements at each time step, performing full state updates only at longer intervals.